PyGameMaker — Tutorial 9: Catch the Coins — Teacher Guide

Companion for the in-app tutorial (**Help > Tutorials > Catch the Coins: Win and Lose**, 5 pages) and the matching student handout and worksheet. Students should have finished Tutorial 2 (events, collisions, score). This is the best first "complete game" lesson: it has a clear win and a clear lose state.

Overview

Students build a game with two endings in four phases: a moving player; coins and an enemy; catching, crashing and score; and a win check. New ideas: **room transitions** (going to a win or Game Over room), **counting instances** (*If count of obj_coin equals 0*), the **project template** "With Game Over Screen", and **Restart game**.

> Ask "how does the game know I won?" The answer, "it counts the coins that are left", is the key idea of the lesson.

Suggested Timing (45 minutes)

The tutorial says 15-20 minutes for a confident student; allow extra time for the template and the win room.

Segment	Time	What happens
Recap and predict	5 min	Show the finished game; ask how a game decides "win" or "lose"
Phase 1: moving player	5-10 min	Pages 1-2 (create the project from the template)
Phase 2: coins and enemy	10 min	Page 3
Phase 3: catching and crashing	10 min	Page 4
Phase 4: winning	10 min	Page 5
Worksheet / exit ticket	5 min	Worksheet parts A-C

Step-by-Step Walkthrough and Common Issues

Phase 1: moving player

- The project must be created with the **"With Game Over Screen"** template; it supplies room_gameover, which the crash event uses later. A project made without it has no Game Over room.
- Three keyboard events on the player, as in Tutorial 2. The player may leave the screen; that is expected.

Phase 2: coins and an enemy

- Coins and the enemy fall straight down (*Set vertical speed 3* in **Create**).

- **Missed coins must come back.** A coin that falls off the bottom of the room is not destroyed: it still exists, far below the screen, so the coin count never reaches 0 and the game becomes impossible to win (in the reference project a missed coin stays alive indefinitely). The tutorial therefore gives the coin and the enemy an **Outside Room** event with *Set variable y = 0*, so anything missed reappears at the top. As a side effect the enemy keeps returning, which keeps the game tense.
- The number of coins placed is the number the player must catch. Five is a good start.

Phase 3: catching and crashing

- Coin collision: *Add to score 1* and **Destroy other instance** (the coin). *Destroy this instance* would remove the player.
- Enemy collision: *Go to room* room_gameover. The room name must match exactly.
- The score is set to 0 in the player's **Create** event and drawn by a **Draw** event on the player.

Phase 4: winning

- A **Step** event on the player checks *If count of obj_coin equals 0*, then goes to room_win. It runs every frame, so as soon as the last coin is caught the win room appears.
- **If there are no coins in the room, the count is 0 immediately** and the game shows YOU WIN! on the first frame. That is the usual "instant win" mistake.
- obj_win_text draws the message at fixed coordinates in a default-sized room (1024×768), and restarts the game on SPACE. The Game Over room from the template restarts the same way.
- The text in the win room is drawn in the default colour (black), which is readable on the bright background the tutorial suggests. On a dark background, use *Set draw color* white.

> **Reference projects.** Download the checkpoint projects from the wiki (09_catch_the_coins_checkpoints.zip): one project for the end of each phase. In the reference project, room_gameover is a small hand-made room, not the template's own. Give a stuck student the previous phase.

Vocabulary Introduced

Term	What it means here
Win state / lose state	The two possible endings of the game
Room transition	Moving to another room with *Go to room*
Instance count	The number of copies of an object that exist, tested with *If count of*
Template	A ready-made project to start from
Step event	Runs every frame; used here to check for a win

Discussion Questions

- How does the game know the player has won?
- What would happen if a coin stayed alive off the bottom of the screen?
- How could we make winning harder? (More coins, faster fall, more enemies.)
- What other games use "collect them all" as the goal?

Differentiation

- **Support:** give the Phase 2 checkpoint; use fewer coins.
- **Extension:** lives; a timer; a second enemy; coins worth different points; a "next level" room instead of the win room.

Worksheet Answer Key

Part A: 1-C, 2-D, 3-B, 4-A.

Part B: 1. obj_player, Collision with obj_coin. 2. obj_player, Collision with obj_enemy. 3. obj_player, Step. 4. obj_coin (and obj_enemy), Outside Room. 5. obj_win_text (and the Game Over room's object), Key press: space.

Part C:

1. The count is already 0, so the game jumps to the win room on the first step (an instant win).
2. No. The missed coin still exists off-screen, so the count never reaches 0. That is why missed coins are sent back to the top.
3. Change *Add to score 1* in the player's **Collision with obj_coin** event to 5 (or use a second coin object with its own value).

Parts D and E: completion and reflection.

Rubric: Catch the Coins

Level	What the game shows
4 - Complete	The player moves; coins and enemy fall and return when missed; catching scores and removes the coin; the enemy leads to Game Over; the score is displayed; catching every coin leads to a win room; SPACE restarts
3 - Working	Moving, catching and crashing work; the win check or the score display is missing
2 - Partly there	The player moves and coins fall, but catching or crashing does not work
1 - Started	The player exists and moves; nothing falls

End-of-Session Checklist

- [ ] Every student has a saved project (Ctrl+S)
- [ ] Every student reached at least Phase 3 (a coin can be caught and the enemy ends the game)
- [ ] Note who needs the checkpoint project next time